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A neural network based digital twin model for the structural health monitoring of reinforced concrete bridges

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ABSTRACT

Developments in Structural Health Monitoring (SHM) research over the past few decades have demonstrated potential in optimising maintenance solutions for degrading infrastructure. The scale of structural deterioration worldwide and the inadequacy of current non-destructive evaluation techniques necessitate the adoption of accessible, quantitative, continuous SHM technology into mainstream asset management practice. This paper seeks to address this significant demand by proposing a robust, end-to-end, fibre-optic sensor (FOS) monitoring prototype which utilises deep neural networks to convert FOS strain output into an interactive digital twin (DT) visualisation. Finite-element validation demonstrated that the prototype was capable of capturing reliable structural analytics, recording an average error of less than 2kNm and an absolute error of less than 0.15 mm for bending moment and deflection respectively. Furthermore, the predictive mean absolute error of the integrated artificial neural network was less than 1με during testing, demonstrating the accuracy of the digital twin when generating baseline strain data for structural analysis.

1. Introduction

Public and private authorities that manage large infrastructure projects have a responsibility and vested interest in maintaining the longevity of their assets. The structural health monitoring (SHM) methodologies which are applied to these projects conventionally prioritise minimal service disruption, efficient resource allocation and a reduction in the degree of operational risk. The value provided by effective SHM systems cannot be overstated. In the United States, there are 625,000 bridges listed by the Department of Transportation, 30% of which have been officially classified as either structurally deficient or functionally obsolete as of 2015 [1]. The ASCE's reporting has directly reflected this, collectively grading US bridge conditions a 'C+' in 1998, 'D' in 2017 and 'C' in 2021 [2]. In 2021, the bipartisan US Infrastructure Investment and Job Act (IIJA) was passed to address the aging state of the built environment, allocating $110 billion USD towards the repair of roads and bridges out of an estimated total of $1.2 trillion in funding [3]. Globally, the cost of corroding infrastructure alone is expected to exceed $1.8 trillion, as a single facet that contributes towards the degradation of assets over time [4].

Furthermore, a BRIME report concluded that across three different European countries – France, Germany and the UK – deficiencies within highway bridges were present at rates of 39%, 30% and 37% respectively [5]. Traditional techniques of non-destructive evaluation (NDE) for the monitoring of these assets have also been severely limited to routine visual examinations which are unable to provide continuous information into the long-term behaviour of these structures. A technical report published by the U.S Federal Highway administration concluded that 95% of element condition ratings provided by state department inspectors vary between two rating points of the average, demonstrating the significant variability of subjective visual inspections [6]. The inaccurate evaluation techniques of current maintenance practices result in economically impractical maintenance schedules and uninformed resource allocation for asset management at an industry scale [7]. The accumulative effects of these stagnant practices and mass structural deterioration pose a significant risk to the built environment, with the potential to produce a marked decline in serviceability over time.

While the term SHM is used to describe a broad domain of data-collection and maintenance practices, in the context of this research SHM is the process of using technology to obtain real-time information to identify and predict possible damages of a building structure over its service life [8]. As outlined by Ko and Ni [9], the main criteria for bridge SHM systems are the:

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- validation of design assumptions and assess structural conditions to provide design precedents for similar structures;
- identification of anomalies in structural response for early damage detection, especially after disasters or extreme events;
- comparison and assessment of structural response over time to allow for optimal maintenance strategies; and.
- collection of substantial amounts of real-time, onsite data to investigate the effects of design actions on structures for the purposes of research.

2. Digital twins for structural health monitoring

Throughout the published literature within the field of Digital Engineering, the term 'Digital Twin' (DT) has been subject to a broad range of definitions. Given the encompassing reach of the field, it is critical for researchers to provide a clear and consistent definition of what a digital twin refers to, both within the context of the proposed innovation and the existing literature. The UK governments Digital Built Britain program defines DT models as *"a realistic digital representation of assets, processes or systems in the built or natural environment"* [10]. More granular definitions of successful DT models emphasise the importance in an ongoing symbiotic connection between digital and physical assets, described by Dang et al. as *"a high-fidelity digital mirror of the physical entity; the former evolves synchronously with the latter throughout their entire life cycle"* [11]. The ongoing development and adaption of DT models with respect to the life cycle of physical assets is specifically relevant to the field of structural health monitoring. In recent years, SHM digital twin models have enabled this feedback between digital and physical assets through the introduction of sensor and signal interrogation systems to measure the current state of the asset. These measurements typically account for some structural metric that industry practitioners conventionally leverage to evaluate structural health, including strain [12], displacement [13] and vibration sensors [14].

In order to clearly define the criteria for a successful digital twin and evaluate the alignment of the proposed DT model to the established literature, a total of eight necessary capabilities have been outlined in Table 1. These key features of conventional DT models were summarised by Ye et al. through consultation with nine experts within the field of structural health monitoring, building information modelling, asset management and digital engineering [15], providing a framework to evaluate the validity of the digital twin solution proposed within this research project.

Table 1
Key criteria for a successful Digital Twin.

Key Digital Twin features	Proposed Digital Twin solution
The model is a *digital replica* of a physical asset.	The model used in the visualization of the post-tensioned footbridge was directly modelled from engineering drawings and specifications, providing an accurate digital representation of the physical asset.
The model is *connected* to the physical bridge, updating based on feedback from the assets current state in near real time.	The proposed solution captures real-time strain data from Fibre Bragg Grating (FBG) sensors that have been embedded within the footbridge during construction.
Data is leveraged as a key tool by the DT model to provide insights into maintenance, monitoring and inspections.	FBG strain data is post-processed by an interrogation system, a range of structural calculations and a deep neural network to generate further structural insights that are visualized within the DT model. These insights provide asset managers additional tools which facilitate informed operational decisions.
The model spans the *entire lifecycle* of the physical asset	As the sensors are installed during construction, data is collected throughout all phases of the asset's lifecycle. Feedback can be recorded from the concrete curing process to the structural response of the asset towards the end of its service life.
There is a *common data environment* which stores all relevant data for the model.	The software developed for this project contains a local database of historic post-processed data which is pulled from to visualize the structural response of the asset. This repository enables the ongoing collection of data within a single, consolidated location for model use.
The digital twin may be used as a *visualization* tool to retrieve data, assisting in communication to stakeholders.	The DT model that has been built within this research project is presented through a custom interactive digital environment, allowing for the retrieval and visualization of historic loading events along with necessary structural metrics.
The digital twin provides *simulation* capabilities which can be physics based or data driven, allowing for the prediction of asset performance	The baseline strain data generated by the deep neural network provides a simulation of the unloaded state of the footbridge. This dataset is used to calculate the delta strain due to loading, capturing the assets structural performance.
The digital twin can *learn* from data to improve future performance	By integrating a deep neural network within the model workflow, the proposed DT model has the capacity to be retrained as the collected datasets continue to grow, allowing for historic data to improve future predictions.

3. Structural health monitoring of a post-tensioned bridge

The novel monitoring strategy that is introduced within this paper aims to provide a holistic tool to record, visualise, and analyse the structural response for a post-tensioned bridge subject to loading events. This solution required the development of a digital twin and deep neural network model, which have been integrated within a combined SHM methodology. Data collected by Fibre Bragg Grating (FBG) sensors are post-processed to visualises the continuous structural behaviour of a post-tensioned footbridge under load. This model-based monitoring strategy allows users to observe abnormalities in the relative strain baseline over time, in addition to the reporting and visualization of critical structural metrics including bending moment and deflection. As a combined monitoring solution, the digital twin prototype and machine learning models introduced within this paper demonstrate the application of fibre optic sensors within an encompassing structural monitoring strategy. The value provided by this integrated digital twin model may be realised through its ability to facilitate informed operational decisions and improve long-term structural integrity, serviceability and resourcefulness over the lifetime of a monitored asset. The broader objective of this research is to improve maintenance and operational services as a precedence of an integrated structural monitoring strategy.

The footbridge bridge itself consists of a 17,900 mm long and 2,600 mm wide reinforced concrete deck superstructure at a 1:20 grade, connected by a movement joint to an adjacent walkway and a construction joint to a mezzanine slab. A rectangular-section concrete column provided a pin support near the midspan of the sloped deck, while the landing slab acts as a fixed-end support. The sensor design consists of three sets of two strain sensor arrays, each containing 16 discrete FBG sensors cast in situ at 1.0 m spacing. Documentation of sensor installation during the construction of the footbridge is provided in Fig. 1. The sensors leveraged for this analysis are attached to the side of the longitudinal reinforcement, with 58 mm of cover between the steel rebar and the top surface of the concrete deck. The top and bottom FOS arrays are installed with a 164 mm difference in depth to capture a holistic strain profile of the footbridge. Additionally, there are two sets of temperature arrays with the sensors attached to the rebars at 5.0 m spacing. The individual FOS arrays attached to the longitudinal rebar are spaced

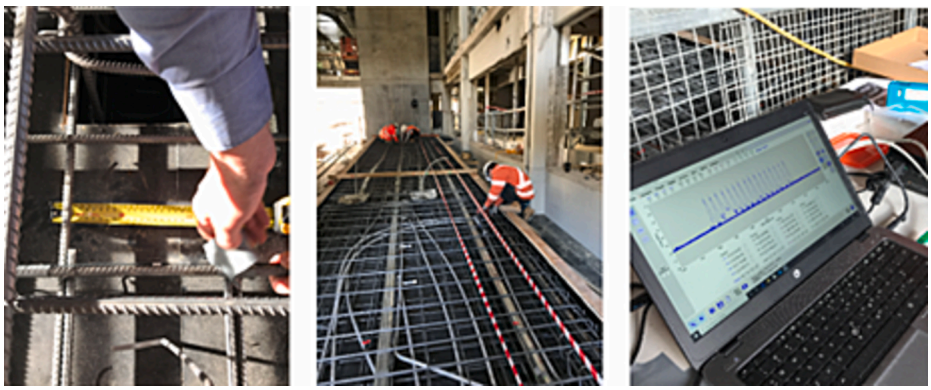


Fig. 1. Installation of FBG sensors during construction of the footbridge.

1080 mm apart. The positioning of the temperature and FBG sensors relative to the longitudinal rebar is further outlined in Fig. 2.

A static Luna Hyperion si255 Optical Sensing Instrument connecting each of these arrays is located in a storeroom underneath the bridge. The output is interpreted by the interrogator which measures 16 FBG wavelengths at 160 nm wide channels to 1 pm precision, equating to ± 1με of total strain or equivalent to a ± 0.1 °C change in temperature. The data is processed via a remote computer connected to an on-site interrogator unit. Concurrently, Python scripts were developed to compile the strain and temperature information for each of the sensors as .csv files which are then stored locally on the computer. The data columns pertaining to each of the strain sensors are labelled with the STR_array_Xm nomenclature where 'array' refers to either A or B and X represents the distance in metres from the rightmost column support. The derivation of the delta strain datasets, deflection and bending response are conducted autonomously, including the use of the Artificial Neural Network (ANN) to generate the relevant baseline data. Derived structural responses are then imported into the Digital Twin environment as a parametric user interface for analysis, as presented in Fig. 7. Finally, finite element modelling used to confirm that the derived metrics agree with the observed data stream. The methodology designed over the course of this research project is illustrated graphically in Fig. 3.

3.1. Fibre Bragg grating (FBG) sensors

Fibre Bragg Grating (FBG) sensors measure the strain and temperature at discrete locations along the bridge based on changes in light intensity in an optical fibre. These FOS are 5–7 mm in size and are comprised of the optical fibre, fibre core and a grating period typically around 500 nm in width. They operate under Bragg's law whereby the spectral properties of the incident light are transformed when it passes through the grating at a particular wavelength [16]. The central Bragg wavelength (λ_b) of the incident light is based on changes in the grating period (Λ) and effective index of refraction (n_{eff})

$$\lambda_b = 2n_{\text{eff}}\Lambda \tag{1}$$

FBG sensors have immense utility in SHM due to their profound ability in accurately measuring the strain and temperature. While most of the sensors on the market are electric-based, FBG sensors are preferred for long-term monitoring due to their suitability to be integrated in highly stressed composite constructions and their superiority in measuring thermal and mechanical effects [17]. FBGs are manufactured as small, compact and lightweight sensors using silica, which is highly immune to electromagnetic interferences and can therefore function well in electrically noisy environments. They are durable and can withstand tensile strains > 10,000 $\mu\varepsilon$ and have the additional capacity for multiple sensors to be multiplexed onto a single optical fibre cable.

The acquisition of FBG strain and temperature information is based on the relative shift in Bragg wavelength signal caused by changes in the effective refractive index and periodicity of the fibre optic cable grating. Deformations from mechanically applied loading can affect this shift but this is not the only factor. FBG sensors are highly sensitive to thermal and chemical effects which is problematic as this can create unpredictability in the observed data stream. The major hurdle is in extracting a true strain measurement after isolating the data fluctuations induced by thermal contraction and chemical resin shrinkage [18]. Therefore, the temperature on the FBG sensors needs to be compensated through considering the refractive Bragg wavelength shift ($\Delta\lambda_b$) as a linear perturbation of the total strain ($\Delta\varepsilon$) and temperature change (ΔT). Upon expressing it in this manner the Bragg wavelength shift is contingent on the strain impact caused by force and the thermal strain varied with

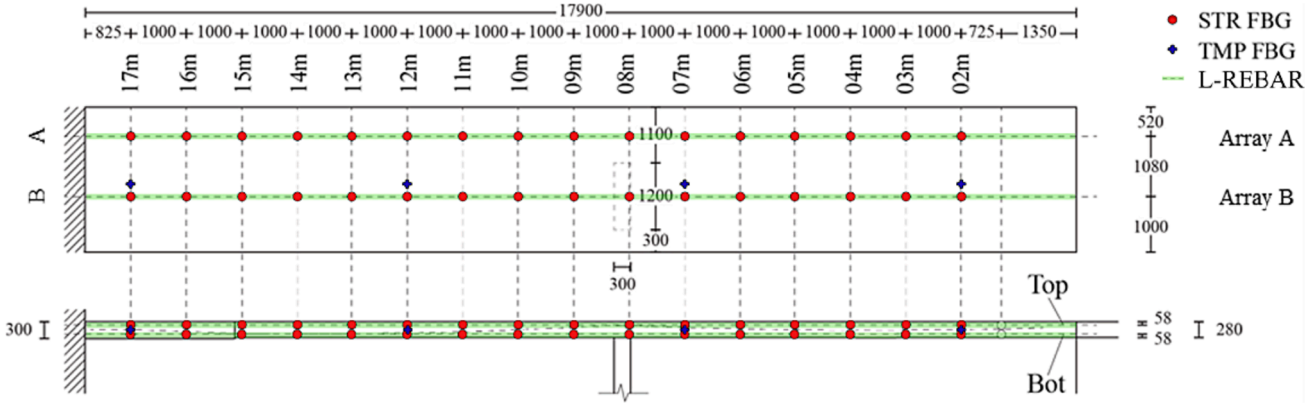


Fig. 2. Plan and section of installed FBG sensor layout.

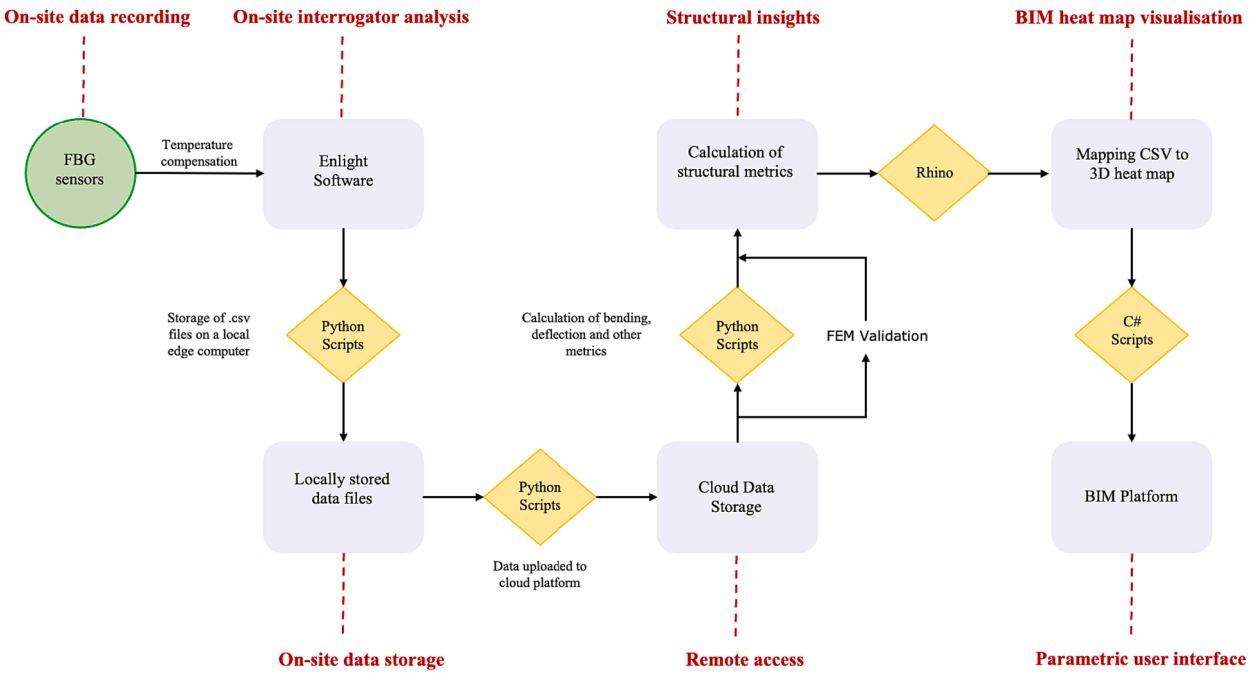


Fig. 3. Data architecture of the proposed digital twin.

temperature,

$$\frac{\Delta\lambda_b}{\lambda_b} = K\Delta\varepsilon + \beta\Delta T \tag{2}$$

where K is the strain-optic coefficient and β is the temperature sensitivity factor of the optical fibre.

3.2. Structural analysis

The post-tensioned bridge monitored within this research project can be simplified as a beam structure. The application of dynamic point loads, such as from pedestrian traffic, will cause the beam to deform in a manner that is dependent on the boundary conditions specific to this structure. As the beam bends, the longitudinal strain varies linearly with the distance from the beam's neutral axis. However, the neutral axis may be shifted in response to temperature variations or if any physical damage such as cracking was present. These structural complexities can be resolved by defining the curvature, κ, as the difference in strain measurements at two arbitrary positions in the cross-section divided by their distance of separation [19].

Two equidistantly placed strain sensors parallel with the neutral axis can be used in this instance to extract the curvature generated from a mechanically induced load:

$$\kappa = \frac{\varepsilon_{\text{bot}} - \varepsilon_{\text{top}}}{h} \tag{3}$$

where ε_{top} and ε_{bot} represent the top and bottom FBG strain sensor readings and h is the vertical distance of separation between their centre of masses. Assuming that the structure is uncracked during most of its operational life, the bridge behaviour can be represented simply as beam elements exhibiting linear elastic deformations. Here the induced bending moments are a component of the flexural stiffness and the deflections can be estimated through the double integral of curvature. In real time, these parameters can be autonomously retrieved through an exact curvature function obtained using regression methods and with respect to the longitudinal bridge span.

$$\kappa = a_n x_n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0 \tag{4}$$

where x is the discrete longitudinal position of a strain sensor along the bridge span and a_0, a_1, …, a_n are the coefficients of a degree n curvature polynomial. For a generic span with n boundary constraints, a maximum $n+1$ degree for the curvature function shall be used for the entire span to avoid any issues with overfitting of the dataset that may arise with the use of higher degree polynomials. The accuracy of the approximated deflection result however is dependent on the closeness of the sensor spacing and improves as more data is collected [20]. To account for this, the optimal degree of the curvature polynomial may be selected according to the optimal R^2 regression coefficient value which reflects the correlation of the function with the dataset [21]. For structural calculations presented within this paper, the modulus of elasticity of the concrete, E_c, was taken as 38 GPa as obtained from concrete testing and for steel, E_s, it was taken as 200 GPa. The reinforcement bars were taken as having a yield strength 500 MPa and the post-tensioned slab was taken to have a compressive strength of 50 MPa.

4. Experimental load testing

The site experiment was conducted nine months after the concrete deck was poured. The primary objective of the experiment was to analyse the behaviour of the FBG system under a known loading condition and to calibrate the embedded sensors. The experimental data would serve to verify the data architecture, whereby a DT would be able to visualise the structural effects of the experimental load consistent with a finite element analysis.

The applied load consisted of two operators guiding a pallet jack carrying 32 weights across the bridge, with a combined mass of 1.1 tonnes. The loading of the footbridge across marked sensor locations is documented in Fig. 4, including the selected testing apparatus. For the purpose of analysis, the load of the crate and operators was considered as a uniform pressure acting over a 2.0 m long × 1.0 m wide area. The locations of the strain-FBG sensors in Array A were marked on the bridge based on the construction drawings. The calibration of the data was performed by moving the load from mezzanine slab end to the walkway landing, with the load centred over each marked sensor at one-minute intervals. The duration of the test was 960 s in total, excluding set-up and movement off the bridge.

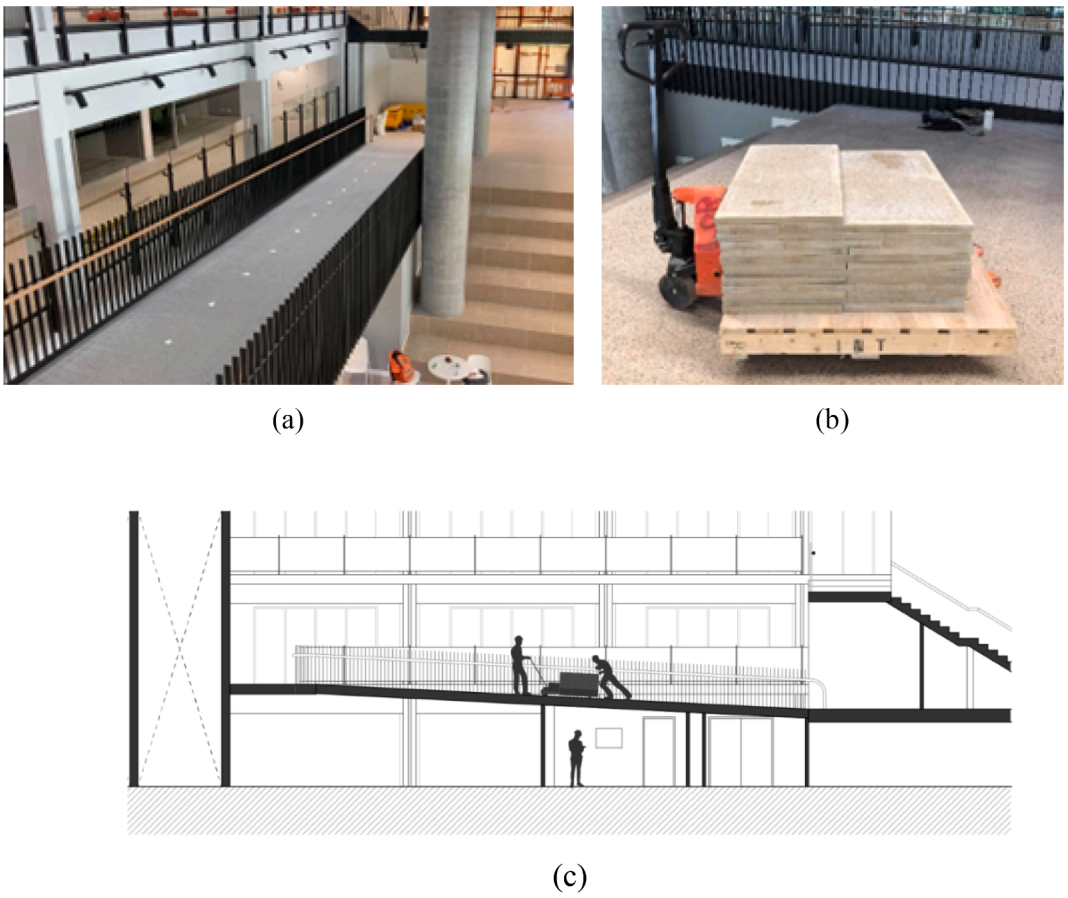


Fig. 4. (a) Footbridge with marked FBG locations, (b) pallet crate load, and (c) section of experiment.

4.1. Initial design considerations

The site experiment was designed to ensure that the derived structural calculations were reliable and that they can be used for subsequent structural analysis. Several design considerations, however, needed to be accounted for in the calibration of these sensors. Firstly, when an FBG sensor is embedded into a structure and is subjected to structural deformation, a strain transfer mechanism is initiated between the optical fibre and host structure. Thermal variations create a prominent structural effect, leading to volumetric changes in the optical fibre and affecting the accuracy of the recorded strain measurements [22]. Therefore, any individual set of strain readings cannot be interpreted directly as the fibre optic cables will be subjected to varying degrees of initial bending due to the construction process and temperature [23]. Instead, the relative change in the strain reading from an original reference level (referred to as the baseline) needs to be considered to accurately represent the structural deformations [24]. Fig. 5 highlights this 'delta strain' (Δε) shift on a sample FBG dataset as initiated by a mechanical loading event.

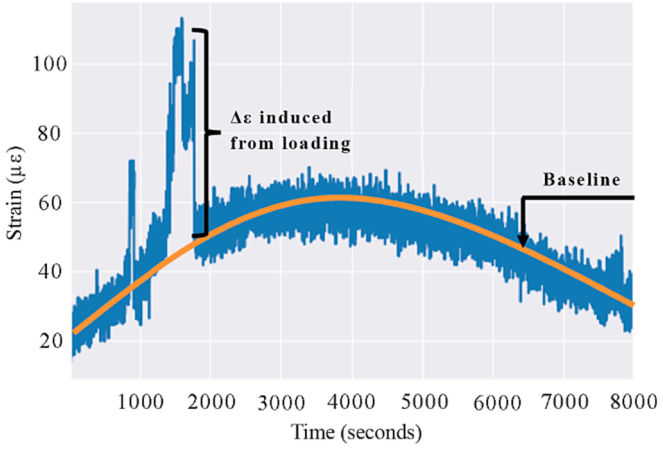


Fig. 5. Relative change in strain measurement induced by a loading event.

4.1.1. Mechanical and thermal strain

Temperature-compensated FBG strain sensors generate total strain measurements which is the summation of thermal and mechanical strain. These total strain readings fluctuate sinusoidally as the structure expands and contracts due to temperature changes. Separation of the mechanical strain is therefore necessary to accurately capture the structural response. This is achieved by taking the coefficient of thermal expansion (CTE), α, of concrete as 10με/°C, or a result determined from test data as specified in AS5100.5:2017 (Clause 3.1.6(a)) [25] and subtracting it from the total strain:

$$\varepsilon_{\text{mech}} = \varepsilon_{\text{total}} - \alpha \Delta T \tag{5}$$

Here $\varepsilon_{\text{mech}}$ and $\varepsilon_{\text{total}}$ represent the mechanical and total strains respectively and ΔT is the change in temperature from a reference value. In this format, the delta strain computations do not require complex modelling algorithms as the general strain behaviour is steady and any load-induced deviations can easily be subtracted from a horizontal baseline. It should be noted, however, that this approach is imperfect as the CTE of Portland cement concrete ranges from about 7.2 to 14.4με/°C depending on the variation of the concrete's component materials [26].

5. Proposed machine learning framework for structural monitoring

In order to generate baseline datasets, an artificial neural network was created to model strain behaviour as influenced by temperature variance. Within the context of the load testing experimentation, the proposed neural network was introduced to generate the baseline strain datasets necessary to derive further structural analytics. The neural network model selection was driven through a comparative study from a broad range of machine learning (ML) models. Strain baseline monitoring is a key element of FBG structural analysis, as it provides a relative framework from which deviations and abnormalities may be observed. The strain output from this analysis has been visualised through a digital twin model, which displays historic structural analytics. A comparative analysis was conducted to evaluate the performance of the proposed neural network solution against results obtained through Linear, Ridge and Lasso regression.

5.0.1. Linear regression

As FBG data collection requires the monitoring of temperature proximate to the installed sensor locations, ML models have been successfully implemented in the past to investigate the relationship between the recorded Bragg wavelength and local temperature. This was directly demonstrated by Sarkar et al., in which thermal variance was identified from strain sensor measurements [27]. For modelling cases in which the dependent and independent variables have an established causal link, simplified linear regression models are a common machine learning solution. This is due in part to their computational efficiency, simplicity, and interoperability of results. Complexity may be introduced to these models if multiple independent variables are used, in which the multivariate linear equation of the model weights may be taken as follows.

$$f(X) = \beta_0 + \sum_{i=1}^{M} X_i \beta_i \tag{6}$$

This equation assumes multiple inputs (X), where M is the number of model input variables, β_i is the coefficient of regression and β_0 is taken as the regression intercept. The coefficient of regression represents the mean change of the dependant variable, strain, and is derived by reducing the sum of squared errors between the predicted strain and the recorded strain data.

5.0.2. Ridge & Lasso regression

Notably, the regression coefficients generated by the least squared method have a high degree of variability which may compromise the validity of the model output. Ridge and Lasso regression techniques comparatively produce biased results with a much lower degree of variance by integrating a penalising regularisation coefficient into the cost function. Specifically, ridge regression utilises a regularisation coefficient to penalise the squared coefficients, illustrated in the following cost function.

$$\sum_{i=1}^{N} (y_i - \widehat{y}_i)^2 = \sum_{i=1}^{N} \left(y_i - \sum_{j=0}^{M} (\beta_j X_{ij}) \right)^2 + \lambda \sum_{j=0}^{M} \beta_j{}^2 \tag{7}$$

The penalisation term, λ, effectively regulates the model coefficient values β, while $\widehat{y}_i$ and y_i represent the predicted and target values respectively. Within this context, X_{ij} represents the value of the *j-th* feature for the *i-th* data point, across a total of N points. As the regularisation coefficient increases, the regression coefficients shrink towards a minimum value. Comparatively, Least Absolute Selection Operator (Lasso) regression utilises a penalisation term to optimise the absolute weight rather than the square of the weights. For Lasso regression, the cost function is defined as follows.

$$\sum_{i=1}^{N} (y_i - \widehat{y}_i)^2 = \sum_{i=1}^{N} \left(y_i - \sum_{j=0}^{M} (\beta_j X_{ij}) \right)^2 + \lambda \sum_{j=0}^{M} |\beta_j| \tag{8}$$

The regularisation techniques within both Ridge and Lasso models serve to prevent overfitting and control the weighted coefficients. Lasso deviates from Ridge regression through its mediation of the absolute penalty function. As the weights are increasingly penalised, Lasso regression shrinks the estimate towards absolute zero, allowing for optimised feature selection and filtering [28].

5.0.3. Artificial neural network

In order to quantify the performance of the proposed Deep Learning framework, a comparative analysis was conducted across all of the tested model. This included the calculation of the mean squared error (MSE) for each of the model's output, where y is the sampled data and $\widehat{y}$ is the model output from N data points. In the evaluation of model performance after the validation phase, it is desirable for the MSE and mean absolute error (MAE) to be as close to zero as possible. Furthermore, in order to test the model's robustness and validity, the distribution of the errors produced must be Gaussian according to the Central limit theorem. Finally, the model performance may also be quantified by calculating the root mean squared error (RMSE) and the coefficient of determination (R^2).

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \widehat{y}_i)^2 \tag{9}$$

$$\text{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \widehat{y}_i| \tag{10}$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \widehat{y}_i)^2} \tag{11}$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \widehat{y}_i)^2 \binom{n}{k}}{\sum_{i=1}^{N} (y_i - \overline{y}_i)^2 \binom{n}{k}} \tag{12}$$

5.1. Proposed digital twin

The heatmap-style meshing process that was designed for this structural monitoring system utilises 50 mm square element created using the parametric visual programming software Grasshopper [29]. Within this Grasshopper script, several number slider control tools provide the ability to tune the parameters that define the generated geometries and analysis, including the element sizing, critical bounds, timestamp selection and gradient colour scheme. An example visualisation isolated from this Grasshopper script is illustrated in Fig. 6, including the meshing, interpolation, and visualisation of the desired structural analysis. These control tools are stored with the exported data in a standardised format, allowing the DT tool to import the heatmap-style visualization and its corresponding defined user tools, outlined in Fig. 7. This structured transfer of information between the developed scripts and programs allows for the continuous generation and interpolation of structural data within Grasshopper, which can be stored for viewing within the DT interface without requiring the recalculation of the parametric visualizations at each timestamp.

5.2. Digital twin prototype

The DT prototype was programmed with the objective of communicating significant quantities of data to provide actionable insights into the structural response of an asset under load. The node structure created within the Grasshopper script allows the interpolated data to be exported as a .csv file, in which the location coordinates and value of each shell element are categorised and recorded. The generated .csv file

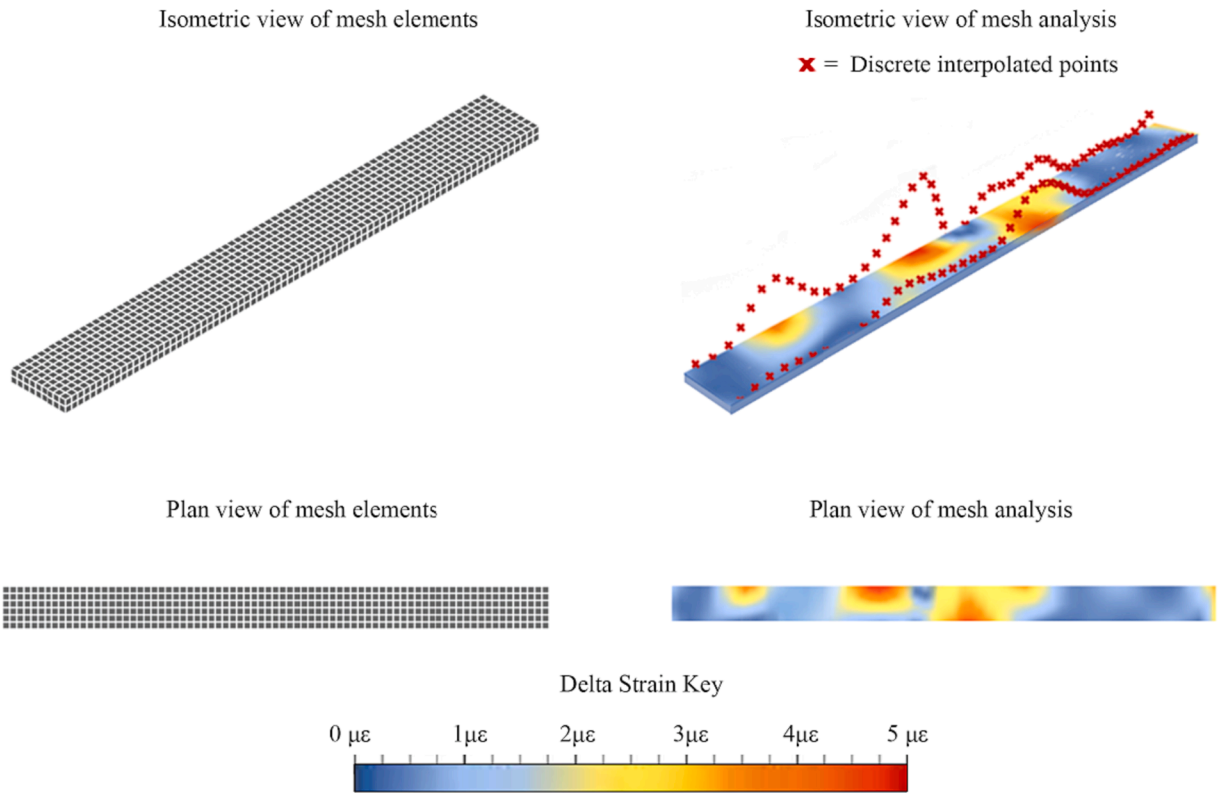


Fig. 6. Strain heatmap generation example.

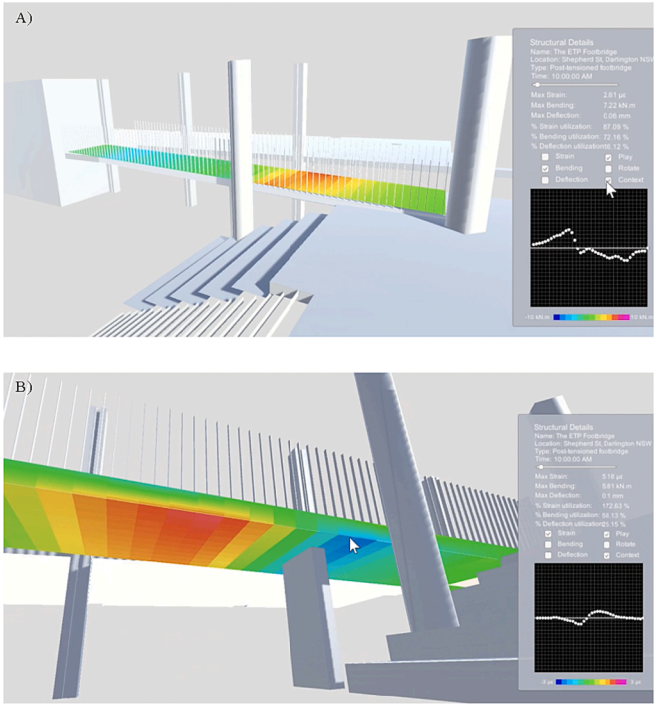


Fig. 7. Digital twin visualisation (A) mesh plane for moment – (B) mesh shell for delta strain.

is then imported into the DT environment, allowing a user to interact with the generate structural data. This external tool was developed to create a SHM workflow in which the user can directly interact with the analysis in an environment that is directly focused on capturing and visualising structural response. The user interface provided within the prototype controls the mode of analysis, the selected timestamp, the critical structural bounds and the animated timeseries. A plotted graph is provided within the UI to directly compare conventional analysis against the heatmap-style visualization, as a holistic overview of the structural response during the loading event (Fig. 7). Critical and maximum values are also summarised within the user tools section, such that the user can identify and quantify the induced strain, bending moment and deflection within the structure.

5.3. Numerical analysis

To validate the observed structural response, the results of the experimental loading condition were compared to a finite element method (FE) model. Model properties and assembly were assigned in accordance with the onsite concrete testing and construction drawings respectively and are given in Table 2 and Fig. 8. The construction joint at the mezzanine slab was modelled as a pinned connection. The concrete deck was modelled as a solid three-dimensional extrusion of a concrete damage plasticity (CDP) material section in Abaqus. The tensile and compressive reinforcement layers were modelled as wire beam elements with solid circular sections and assembled as per the construction

Table 2
Abaqus model parameters.

Part	Concrete deck	Longitudinal reinforcement	Stirrups	Post-tensioning duct
Model	3D solid brick	Wire	Wire	Wire
Dimensions (mm)	2600 × 280 × 15200 2600 × 300 × 2700	ϕ 10, ϕ 12, ϕ 16	ϕ 16	49.1 × 10
Approx. mesh sizing (mm)	70	70	70	70
Mesh method	C3D8R	B31	B31	B31

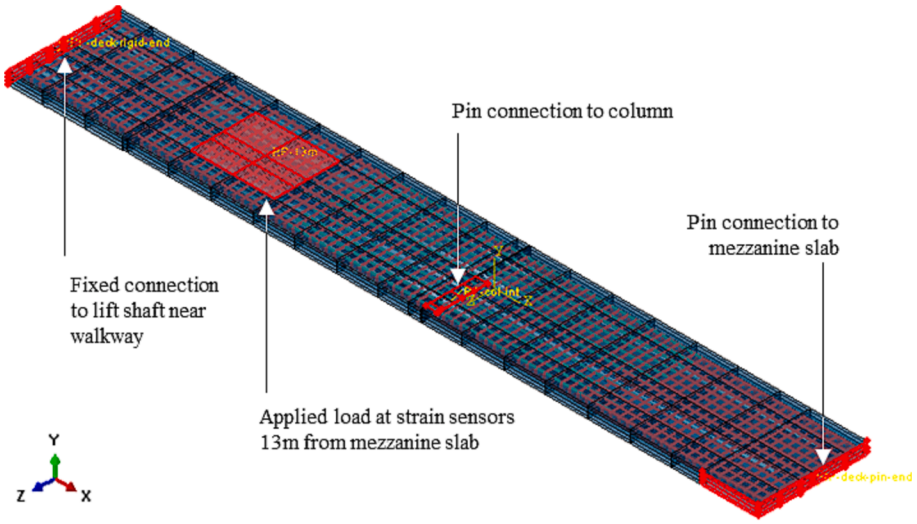


Fig. 8. FE model assembly, partitioning, and boundary conditions.

drawings, accounting for the changing section geometry. As post-tensioning tendons provide negligible bending stiffness, the post-tensioning ducts were modelled as thin flat rectangular sections of area equivalent to a ϕ25mm solid circular bar. Geometrically, flat bars provide axial stiffness while minimising the *I*-value [30]. All reinforcement was modelled as embedded in the concrete with a 'rough' no-slip condition, reflecting the effects of the threaded reinforcement. The pallet crate was applied as a point load acting over a 2.0 m long × 1.0 m wide rigid body, corresponding to the assumed application area in the site experiment. A pin boundary condition was applied to the load surface by a reference point. For this FE study, a pin (translation fixed, rotation free) and a rigid (translation and rotation fixed) boundary condition was applied over the entire surfaces of the mezzanine and walkway landing ends respectively. A pin boundary condition was also applied over a 1200 mm × 300 mm area at the bottom surface of the bridge to model the effects of the central column support.

A convergence study was undertaken using an *h*-refinement framework [31] to test the validity of the meshing method, the results of which are outlined in Table 3. A given three-dimensional brick element is divided into eight elements, as divisions are made in each axis. For coarse meshes where partitioning would dramatically distort the aspect ratio (AR) of the brick elements (beyond AR = 2), the partitions at the reinforcement were removed, however the column placement and shape was preserved to maintain a consistent boundary condition. The mesh was subdivided equally across all elements along the length of the bridge. A local refinement methodology was considered for the mesh convergence study, as refinement of elements about the area of stress concentration improves accuracy of the results without dramatically increasing the processing time [32]. Refinement was conducted using a coarse mesh equal to the depth of the section with refined meshes constructed until the failure from the processing system at $h = 0.0175$. By calculating and plotting the natural logarithms of average mesh size, h, against relative change in deflection, Δu_y, the mesh convergence rate was derived. From this analysis, the convergence rate (CR) was determined to be 4.36, which implies the mesh was made approximately four times as accurate with every further refined mesh.

Table 3
h-refinement calculations.

h (m)	Elements	u_y(m)	Δ (m)	ln*h*	lnΔ
0.280	4388	−0.00402		−1.27	
0.140	11,998	−0.00028	0.00374	−1.97	−5.58
0.070	50,780	−0.00023	0.00005	−2.66	−9.90
0.035	341,520	−0.00022	0.00001	−3.35	−11.64

6. Results

6.1. Model performance comparison

In order to evaluate the effectiveness of this end-to-end structural monitoring solution, the results comparison has been divided into two distinct categories: the validation of the various machine learning (ML) models for baseline generation, and the validation of the digital twin prototype which leverages this baseline information. The initial theoretical observation of a linear relationship between the Bragg wavelength and local temperature was useful to identify potentially effective machine learning models for testing. However due to the stochastic nature of ML prediction, the variance induced by hyperparameter tuning and the disparate functionality of the chosen models, it is not possible to extensively evaluate these models a priori. A critical review of the tested machine learning techniques is therefore necessary in order to identify the most robust and accurate predictive solution. The performance metrics are summarised in Table 4 for the testing and training of the selected machine learning models.

Further residual analysis was conducted to extend the performance analysis for the tested models, outlined in Fig. 9. Models which are systematically biased towards under- or over-prediction will typically exhibit a degree of skewness in the distribution of their errors, such that the model inadequately describes a particular trend that is present within the data. The density distribution histogram provided in Fig. 9 plots the model prediction errors for each of the machine learning models.

The error density plots provided in Fig. 9 allow for a direct comparison of the various model residual distributions, in which the idealised Gaussian fit is superimposed as a red curve centred around a mean of zero. The kernel density estimate curve (KDE) represents a non-parametric probability density function that estimates the smoothed shape of the model's residual distribution, generated using the following equation:

$$\widehat{p}_n(x) = \frac{1}{nh}\sum_{i=1}^{n} K\left(\frac{X_i - x}{h}\right) \tag{13}$$

where h is defined as a smoothing bandwidth that controls the degree of smoothing that the curve is subject to, X_i is the smoothed data point and $K(x)$ is the Gaussian kernel function. It follows that for each of the tested machine learning models plotted in Fig. 9, the central limit theorem is satisfied if the black KDE curve is proximate to the red theoretical

Table 4
Statistical metrics of machine learning models.

Model	Training Data				Test Data			
	MAE	MSE	RMSE	R^2	MAE	MSE	RMSE	R^2
ANN	1.096	1.923	1.387	0.9960	0.732	0.845	0.919	0.9959
Linear	1.144	1.957	1.399	0.9907	1.143	1.949	1.396	0.9908
Lasso	1.147	1.961	1.401	0.9907	1.133	1.931	1.390	0.9908
Ridge	1.199	2.152	1.467	0.9898	1.203	2.155	1.468	0.9897

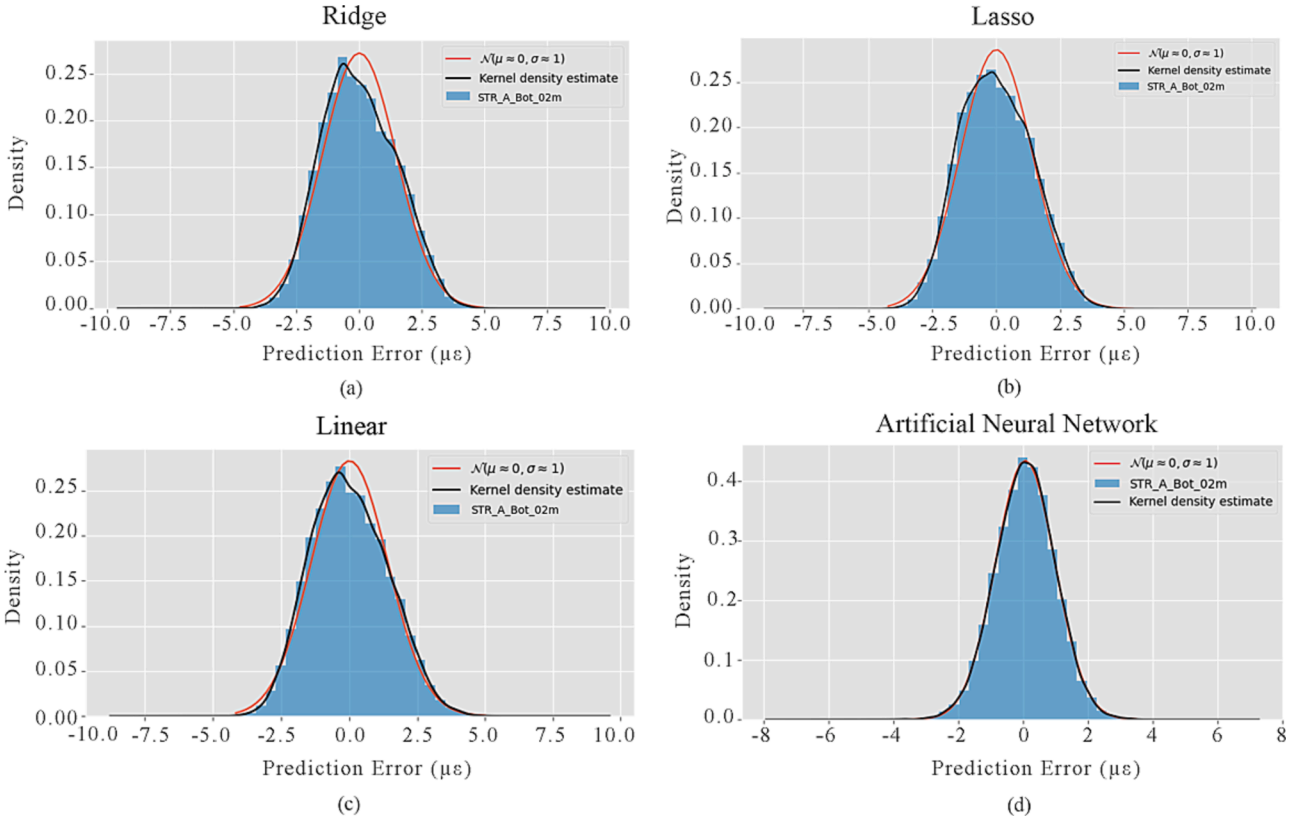


Fig. 9. Prediction error density plot for each model.

Gaussian distribution curve. Given this performance criteria, the statistical analysis graphically demonstrates that the residuals produced by the ANN are more normally distributed and homoscedastic relative to the other machine learning techniques that were tested. In contrast, the Ridge, Lasso, and linear techniques exhibit a degree of skewness and asymmetry relative to their target values. It is therefore necessary to conduct further statistical analysis to identify the degree to which the model errors satisfy the central limit theorem for each of the selected ML techniques. This observation aligns with the previous statistical analysis and the conclusions drawn from the preliminary performance metric comparison, demonstrating that the ANN is an appropriate candidate to be used in conjunction with the DT prototype. In order to validate this conclusion further, quantile–quantile (Q-Q) plots of the various modelling techniques are provided in Fig. 10. For this analysis, the theoretical quantile samples are defined by a Gaussian distribution. A 45-degree red line is superimposed as a guide to demonstrate the case in which the theoretical and sampled quantiles come from identical distributions. Deviation from this theoretical target line signify a departure from the theoretical normal distribution, which is undesirable when selecting an appropriate machine learning model.

The Q-Q plot provided in Fig. 10 demonstrates that the ANN residuals are drawn from a symmetric normal distribution due to the linear alignment of the sampled quantiles to the theoretical target line. Deviations from this line are minimal, indicating that the residuals of the ANN are acceptably Gaussian in nature. In comparison, the tails of the Ridge, Lasso and linear Q-Q plots are lightly curved away from the perfect fit, demonstrating that the distribution of their respective prediction errors is less satisfactory. While a degree of deviation is common when visualising ML Q-Q plots, graphs which are excessively "S" shaped indicate that the distribution of the model errors is influenced by skewed prediction data. The extent to which the tails of the Q-Q plot deviate is indicative of the degree to which the model residuals differ from the expected Gaussian distribution.

6.2. *Validation of digital twin prototype*

The predictive output from the ANN generated the baseline datasets necessary to compute the structural metrics visualised by the DT. Through an analytical study of the induced delta strain measurements caused by the load testing experiment, validation of the key structural metrics in the format of deflections and bending moments was therefore possible. The critical loading case was chosen when the footbridge experienced the most prominent structural deformations, specifically the instance whereby the pallet jack was equidistant between the mid-span supporting column and fixed end support (t = 13 min). With reference to the defined sensor layout, this was located approximately at

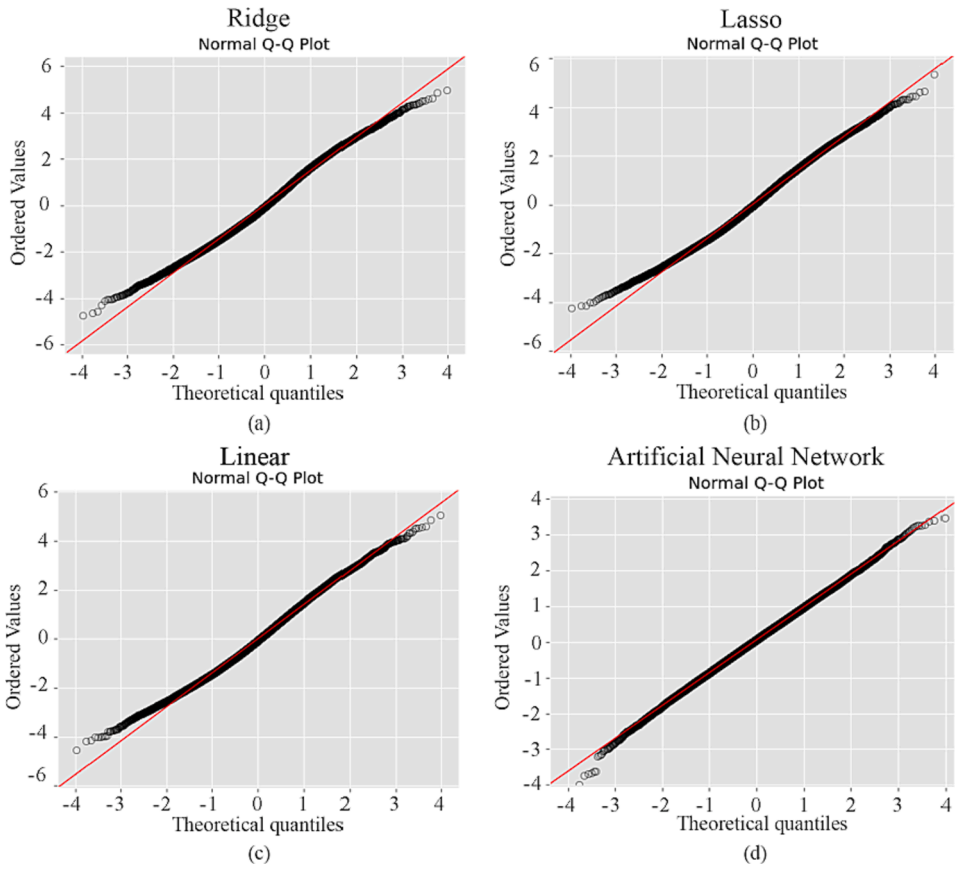


Fig. 10. Normal Q-Q plot of residuals for each model.

the strain sensors embedded 13 m from the mezzanine slab. Results obtained from experimental loading were averaged per minute interval to account for the influence of signal noise. In addition to bending moment, the real-time vertical deflections of the bridge were also analysed and, at most, a degree-4 regression polynomial was considered to avoid curve overfitting. The optimum polynomial was selected based on the highest coefficient of determination R^2 after cycling through each of the degrees up to and including 4. The absolute error was used as an indicator of the DT performance used and to compare the outputs generated between the DT model and FE analysis. These results have been summarised in Table 5 and 6.

7. Discussion

7.1. Machine learning baseline

From the statistical results summarised in Table 4, the ANN performed the best across all measured evaluation metrics for both training and testing datasets. Notably, the ANN had the smallest MAE, MSE and RMSE, indicating that the ANN model was capable of generating baseline strain datasets more accurately and consistently relative to the other tested modelling techniques. Furthermore, the coefficient of determination for the ANN was measurably greater than that of the other machine learning models. In contrast, the Ridge regression model performed the worst across all measured statistical metrics, indicating that the Ridge modelling technique was less effective in capturing and generalising the relationship between strain and temperature. The accuracy of the neural network's predictions is further demonstrated in Fig. 11, in which the model's output is consistently within the threshold of sensor noise present within the test data, which is approximately ± 1με for the installed fibre optic sensors.

Notably, the performance metrics of the linear and Lasso regression models are extremely proximate to one another. Conceptually, this is due to the fact that the Lasso modelling technique is a modification of the linear method, in which the model weights experience a penalisation controlled by the regularisation coefficient (λ). That is, if the hyperparameter λ is taken to be zero, the Lasso model would produce the same coefficients as the linear model for the same given dataset. During the hyperparameter tuning phase, the optimised regularisation coefficient was tuned down to a value of 0.001, significantly reducing the penalisation to which the model weights are subjected. Hence, within the Lasso model environment, the performance was iteratively improved as the model more closely resembled the linear regression method. It follows that the performance variance between the linear and Lasso models is primarily due to the minimal impact of the tuned regularisation coefficient and the stochastic nature of the tested ML models. Further research would be required to demonstrate the feasibility of alternative Lasso hyperparameter tuning techniques in achieving measurable performance improvements over linear regression modelling.

The efficacy of the footbridge DT was assessed on its ability to replicate the structural deformations induced by loading to a high level of precision. FE modelling was used in this validation to determine the expected numerical values for deflection and bending moment for the loading test. Overall, the post-processed longitudinal bending moments were mostly consistent with FEM recording an average absolute error of less than 2kNm, with the exception of strain sensors in close proximity to the boundary constraints. This is also true for the deflection calculations measuring a negligible highest absolute error of only 0.15 mm. While the overall alignment of the analytical computations with FE method results is promising and demonstrates reliability in the proposed processing algorithms, the system's inability to account for the discrepancies near to the fixed end remains a challenge (>7kNm absolute error) and warrants further review in future investigations. It is hypothesised that the cause of this error is due to the decision not to model the

Table 5
Comparison of Digital Twin and Finite Element bending moment results.

Array	Bending moment (kNm)															
	Str_02m	Str_03m	Str_04m	Str_05m	Str_06m	Str_07m	Str_08m	Str_09m	Str_10m	Str_11m	Str_12 m	Str_13m	Str_14m	Str_15m	Str_16m	Str_17m
DT-A	0.06	0.19	0.03	−0.71	−1.52	−2.86	−5.02	−3.96	−0.94	2.24	5.11	6.69	7.56	3.94	0.82	−1.69
FE-A	0.80	0.53	0.26	0.25	0.25	−0.54	−0.94	−4.89	0.42	4.05	7.78	7.78	6.71	1.79	−3.81	−9.27
Error	0.74	0.34	0.23	0.96	1.77	2.32	4.08	0.93	1.36	1.81	2.67	1.09	0.85	2.15	4.63	7.58
DT-B	0.17	0.32	0.28	−0.52	−1.52	−2.51	−2.93	−3.83	−0.89	1.95	5.02	7.03	6.47	2.95	0.75	−1.81
FE-B	0.80	0.53	0.26	0.25	0.25	−0.54	−0.94	−4.89	0.42	4.05	7.78	7.78	6.71	1.79	−3.81	−9.27
Error	0.63	0.21	0.02	0.77	1.77	1.97	1.99	1.06	1.31	2.1	2.76	0.75	0.24	1.16	4.56	7.46

Note: DT-A and FE-A denotes Digital Twin and Finite Element respectively for strain Array A.
Note: Absolute error taken between DT and FE results.

Table 6
Comparison of Digital Twin and Finite Element deflection results.

Array	Deflection (mm)															
	Str_02m	Str_03m	Str_04m	Str_05m	Str_06m	Str_07m	Str_08m	Str_09m	Str_10m	Str_11m	Str_12m	Str_13m	Str_14m	Str_15m	Str_16m	Str_17m
DT-A	0.04	0.06	0.07	0.08	0.08	0.07	0.05	0.03	0.00	−0.05	−0.09	−0.01	−0.11	−0.09	−0.07	−0.03
FE-A	0.01	0.03	0.03	0.03	0.02	0.01	−0.03	−0.08	−0.14	−0.20	−0.22	−0.21	−0.20	−0.15	−0.08	−0.02
Error	0.03	0.03	0.04	0.05	0.06	0.06	0.08	0.11	0.14	0.15	0.13	0.2	0.09	0.06	0.01	0.01
DT-B	0.03	0.04	0.06	0.06	0.06	0.05	0.04	0.02	0	−0.04	−0.07	−0.09	−0.08	−0.07	−0.05	−0.03
FE-B	0.04	0.06	0.06	0.06	0.05	0.04	0.00	−0.05	−0.12	−0.18	−0.21	−0.19	−0.18	−0.14	−0.07	−0.02
Error	0.01	0.02	0	0	0.01	0.01	0.04	0.07	0.12	0.14	0.14	0.1	0.1	0.07	0.02	0.01

Note: DT-A and FE-A denotes Digital Twin and Finite Element respectively for strain Array A.
Note: Absolute error taken between DT and FE results.

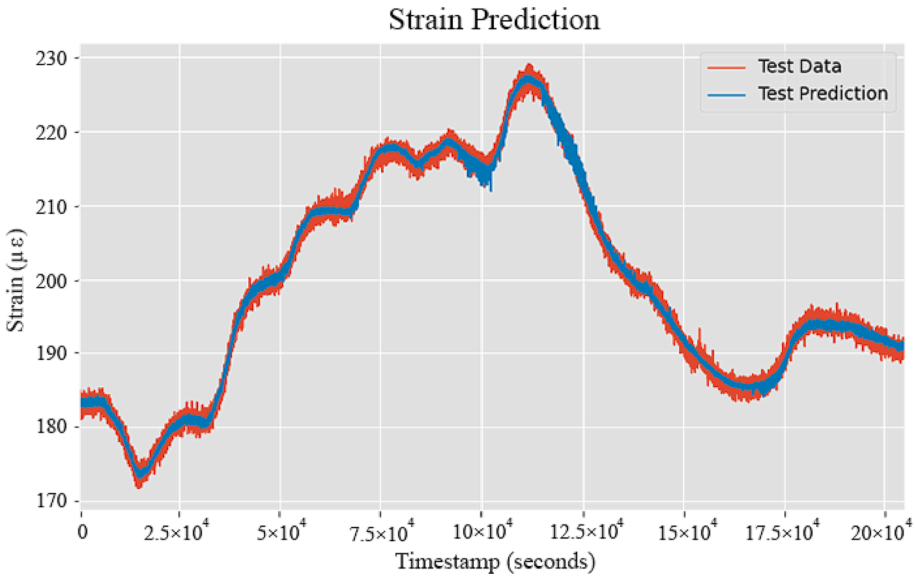
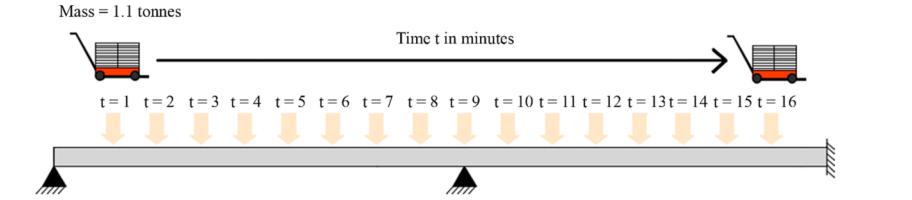


Fig. 11. Artificial neural network baseline strain prediction.

complex three-dimensional effects of the adjoining platform, which would likely induce additional spring stiffness. This was outside the scope of this project, which focused primarily on two-dimensional structural analysis within the digital twin.

These analytical results were also represented dynamically showing a general agreement with FE results and the DT output which was represented in a heatmap-style visualization. The bending moments were mapped to a 2D geometric plane using 16 × 2 cell values (each corresponding with a distinct sensor location) to visualise the real-time structural response. Fig. 12 illustrates this dynamic map for bending moment which clearly demonstrates the movement and corresponding deformations of the bridge during the loading experiment.



t = 1	0.90	0.25	-0.20	-0.18	0.05	0.81	1.94	1.12	-0.08	-0.53	-1.53	-1.66	-1.42	-0.85	-0.30	0.10	Array A
	0.67	-0.44	-0.16	-0.21	0.12	0.60	1.36	1.12	0.02	-0.46	-1.12	-1.29	-1.28	-0.98	-0.65	0.07	Array B
t = 2	1.57	0.90	0.23	-0.36	-0.74	0.14	1.85	1.02	-0.27	-1.29	-1.55	-1.93	-1.96	-1.40	-0.46	-0.50	Array A
	1.16	0.22	0.18	-0.43	-0.38	0.24	0.98	0.71	-0.64	-0.97	-1.64	-1.52	-1.60	-1.27	-0.90	0.09	Array B
t = 3	0.89	1.30	1.64	0.33	0.01	0.34	1.39	0.70	-0.17	-1.01	-1.51	-1.69	-1.62	-0.54	-0.32	0.70	Array A
	0.62	0.86	0.97	0.54	0.22	0.33	0.86	0.72	-0.43	-1.00	-1.40	-1.64	-1.45	-0.92	-0.52	0.53	Array B
t = 4	-0.29	0.45	2.14	1.85	0.98	0.49	0.84	0.47	-0.24	-1.28	-1.75	-1.91	-1.66	-1.20	-0.37	0.52	Array A
	-0.15	0.20	1.36	1.70	1.30	0.63	0.71	0.45	-0.56	-1.17	-1.42	-1.70	-1.36	-0.72	-0.43	0.88	Array B
t = 5	-0.41	-0.56	1.24	2.26	2.51	1.35	0.59	0.22	-0.51	-1.28	-1.67	-1.59	-1.89	-1.13	-0.72	0.43	Array A
	-0.38	-0.24	0.35	2.23	2.43	0.94	0.21	0.33	-0.48	-1.49	-1.69	-1.72	-1.39	-1.29	-0.60	0.35	Array B
t = 6	-0.64	-1.03	-0.06	1.27	2.66	1.99	0.42	0.35	-0.97	-1.40	-1.99	-2.31	-1.85	-1.05	-1.01	-0.07	Array A
	-0.63	-0.68	-0.42	1.36	2.59	2.12	-0.05	0.09	-0.66	-1.45	-2.01	-1.95	-1.85	-0.97	-0.60	0.13	Array B
t = 7	-0.48	-0.53	-0.67	0.38	1.11	2.11	-0.25	0.84	-0.49	-1.22	-1.82	-1.16	-2.01	-1.30	-1.20	-0.24	Array A
	-0.35	-0.38	-0.33	0.09	1.42	1.56	0.00	0.67	-0.30	-0.81	-1.67	-1.79	-1.44	-1.42	-0.99	-0.22	Array B
t = 8	0.05	-0.15	-0.35	-0.21	0.35	0.98	-0.46	1.35	0.79	-0.31	-1.02	-1.29	-1.53	-1.35	-0.81	-0.25	Array A
	0.18	-0.35	-0.12	-0.11	0.50	0.75	0.17	1.42	0.57	-0.16	-0.90	-1.46	-1.42	-0.92	-0.94	0.04	Array B
t = 9	0.10	-0.07	-0.13	-0.54	-0.57	-1.22	-2.05	1.52	2.75	1.79	0.49	0.76	-0.72	-0.83	-1.16	-0.59	Array A
	-0.05	0.31	-0.13	-0.51	-1.08	-1.37	-1.50	1.04	2.69	1.82	0.53	-0.44	-0.91	-0.81	-1.08	-0.51	Array B
t = 10	0.32	0.04	-0.16	-0.51	-1.48	-2.72	-4.49	-0.95	3.09	4.02	3.79	1.78	0.49	-0.31	-1.01	-1.27	Array A
	0.05	0.43	0.22	-0.40	-1.79	-2.18	-3.10	-0.86	2.81	3.73	3.27	2.12	0.38	0.16	-1.17	-1.10	Array B
t = 11	0.30	0.38	-0.05	-0.85	-1.75	-3.40	-5.91	-2.85	2.26	5.27	5.96	4.31	2.74	0.81	-0.87	-1.58	Array A
	-0.11	0.54	0.31	-0.77	-1.84	-2.70	-3.53	-2.85	1.85	4.61	5.30	4.36	2.43	0.57	-1.04	-1.43	Array B
t = 12	0.06	0.30	0.23	-0.76	-1.65	-3.50	-5.70	-4.15	0.38	3.99	6.78	7.01	5.17	2.10	-0.35	-1.71	Array A
	0.36	0.11	0.32	-0.81	-2.06	-2.79	-3.79	-3.74	0.51	3.64	6.30	6.22	4.80	2.06	-0.10	-1.65	Array B
t = 13	0.06	0.19	0.03	-0.71	-1.52	-2.86	-5.02	-3.96	-0.94	2.24	5.11	6.69	7.56	3.94	0.82	-1.69	Array A
	0.17	0.32	0.28	-0.52	-1.52	-2.51	-2.93	-3.83	-0.89	1.95	5.02	7.03	6.47	2.95	0.75	-1.81	Array B
t = 14	0.14	0.17	0.11	-0.79	-1.27	-2.41	-3.97	-3.22	-1.30	0.94	3.04	6.08	7.78	4.91	1.96	-0.94	Array A
	0.24	0.28	0.25	-0.55	-1.45	-2.05	-2.65	-3.37	-1.41	0.69	3.09	5.75	6.66	4.35	2.05	-1.27	Array B
t = 15	0.16	0.27	-0.10	-0.22	-0.69	-1.13	-1.73	-2.11	-0.83	-0.46	0.90	2.69	4.44	5.23	4.37	1.80	Array A
	0.25	0.28	-0.19	-0.33	-1.00	-0.74	-0.93	-1.96	-1.00	-0.21	1.33	2.12	4.22	4.44	4.10	0.46	Array B
t = 16	-0.19	0.37	0.07	-0.36	-0.36	-0.28	-0.09	-0.63	-0.74	-0.16	-0.12	0.45	1.38	2.36	3.92	1.42	Array A
	-0.19	0.37	0.07	-0.36	-0.36	-0.28	-0.09	-0.63	-0.74	-0.16	-0.12	0.45	1.38	2.36	3.92	1.42	Array B

Fig. 12. Digital twin bending moment map.

8. Conclusions

According to the calculated delta strain, deflection and bending moment datasets, the DT solution provided a comprehensive summary of the structural response to experimental loading in the footbridge. Notably, the prototype errors present within the delta strain calculations were primarily influenced by signal noise and baseline accuracy, while the bending moment and deflection calculations for the DT and FE analysis were also subject to a series of structural, material, geometric, and boundary condition assumptions. Despite this potential variability, the critical values of strain, deflection and bending moment derived by the DT program were sufficiently proximate to the output of the FE model, indicating the validity of the solution as a supplementary SHM tool. These results indicate that the monitoring strategy outlined in this paper warrants further research into the application of FOS systems for SHM.

Furthermore, across all of the evaluated performance metrics, the ANN designed within this project produced strain baseline data sets that were more accurate, robust and consistent than the Linear, Lasso and Ridge ML models. These results accumulatively demonstrate that the ANN was the most capable ML technique used to generalise relationships and trends present within the discrete FBG strain and account for any subtle non-linear response that was induced within the structure. With respect to all eight of the key criteria for successful digital twin models derived from the available literature, the proposed deep learning DT model provides an innovative application of digital twinning for the applied purpose of structural health monitoring. As a solution specifically design for the monitoring of post-tensioned bridge assets, the DT prototype and ML models introduced within this paper demonstrate the application of FOS within a holistic SHM strategy. The value provided by this novel DT model may be realised through its ability to facilitate informed operational decisions and maintenance procedures, in turn improving long-term structural integrity, serviceability and resourcefulness over the lifetime of a monitored asset.

8.1. Research limitations

While the novel data architecture has been successful in reproducing the FBG strain and temperature information into practical structural metrics for SHM, it remains limited in its design and functionality, specifically:

- The embedment of fibre optic cables in post-tensioned concrete structures is an imperfect procedure. Stretching and bending of the cables during installation affects the magnitude of the Bragg wavelength recorded by the interrogator. The degree to which this variance influences the proposed strategy will vary between projects.
- The rate to which any specific structural element expands or contracts depends on a multitude of factors including the ambient temperature, humidity and presence of direct solar radiation. For outdoor structures the thermal variation can be significant and should therefore be listed as a scope for further research.
- The efficacy of the structural approach relies on curvature-based rotations. The assumptions and calculations outlined in this paper are suitable for post-tensioned simply supported bridge structures. These assumptions will need to be reassessed for other bridge geometries such as arch, suspension or steel bridges.
- The simplification of the bridge structure that was used to improve the speed of structural calculations will not be fully applicable for other types of structure. The elimination of the 1:20 grade and generalization of the structure's boundary conditions were contributing factors towards the discrepancy between the DT model output and finite element calculations, which were justified within the scope of this research project. These will need to be assessed on a case-by-case basis.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: S. A. Hadigheh reports financial support was provided by New South Wales (NSW) Smart Sensing Network (NSSN). S.A. Hadigheh reports financial support was provided by Laing O'Rourke. Author (T. Hielscher) and Co-author (S. Khalil) previously employed at Laing O'Rourke.

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